INSTITUT D'ENSEIGNEMENT SUPÉRIEUR DE RUHENGERI

COMBO_A3_G03_REPORT

AgriSoko Pocket G03 — UI/UX and AI system report

Group 03 · Verification COMBO-G03-7014 · Version 1.0.0
Repository https://github.com/Mogadi/agrisoko-pocket-g03

1. Problem and users

AgriSoko is preparing a short demonstration for potato cooperatives, employers and innovation partners in Musanze. Visitors needed more than a running APK: they needed to know *who* built each part and how to verify it. AgriSoko Pocket G03 lets a person ask a natural question and receive a profile that only repeats approved facts.

Product promise

Ask by name, nickname, role, skill or feature. See confidence, the reason for the match, an evidence-based introduction, and working links to GitHub. When the network is gone, exact identity lookup still works and is labelled offline.

#	Member	Reg. no.	Primary role
1	Mohammed Osama Hasan	25/27014	Product and GitHub lead
2	Nuzha ZainEl-Abdeen Mohammed Ismail	25/27419	AI and data engineer
3	Ehab Fakhralden Mohamed Hamid	25/27950	API and integration engineer
4	Mojtaba Abdalitieef Ahmed	25/27660	Mobile UI and navigation engineer
5	Lazarus Simboya Ira Inyasio	25/28180	Offline and quality engineer
6	Feras Saifaddin Ismail Mohammed	25/27916	Release and evidence engineer

Mohammed Osama Hasan is first on this product as group leader and Product/GitHub lead. No cooperative sales metric is claimed on this page.

2. Information architecture

Tabs: Home · Ask · Team · About
Stack: Result (AI match / no-match / error) · Profile

Online flow

1. User types a question. Empty and over-long text is rejected on the device.
2. NetInfo reports online. Axios POST /ask with an 8-second timeout.
3. Result shows member, confidence, reason, introduction, source badge “Live match”.
4. Profile opens evidence URLs with Linking.

Offline flow

1. NetInfo reports offline. The phone does not call TF-IDF.
2. Exact name, nickname or role is compared to the bundled directory.
3. A hit is labelled “Offline match”. A miss explains the limitation and lists names.

Recovery

Timeout and 5xx/connection errors show Retry and keep the last useful introduction on screen. Cached directory and last ask live in AsyncStorage.

3. Annotated interface — Home, Ask, Result, Directory



Type sizes start at 16px body and 13px labels so large-text settings remain readable. Primary green on cream meets contrast for body copy. Interactive controls have accessibility labels.

4. Annotated interface — Profile, no-match, error, offline

PROFILE
Contribution, skills, measured evidence, GitHub links with failed-link alert

NO MATCH
Plain refusal, suggestions of names/roles/features, no invented person

TIMEOUT / SERVER
Retry button, last useful result kept, no crash

OFFLINE
Amber “Offline match” badge; feature questions without a name are refused

About screen (not pictured here) holds data source, vectorizer text, fingerprint, limitations, GitHub URL, APK version, verification code and a writable API base URL.

5. AI design

One searchable document per member is built from name, aliases, GitHub username, role, skills, target user, problem, contribution, feature, outcome metric, tester feedback and introduction draft. Empty optional fields are skipped; the member stays in the index.

Choice	Rule
Vectorizer	FeatureUnion of word n-grams (1–2) and char_wb n-grams (3–5), sublinear TF, L2
Similarity	Cosine
Exact boost	+0.25 when a normalised name, alias, GitHub username or name token (>2 chars) is contained in the query; cap 1.0
Threshold	0.45. A change must be justified by evaluation/queries.json.
Grounding	introductionDraft and evidenceLinks copied verbatim. No generative API.

Fingerprint: SHA-256 of canonical members.json, exposed as sha256:… on /model-info.

6. API and integration

Endpoint	Contract
GET /health	status, version, groupCode, modelVersion, memberCount
GET /members	Safe directory fields only
GET /members/{id}	Full approved profile or 404
POST /ask	{query} → match, confidence, reason, introduction, evidence, modelVersion
GET /model-info	vectorizer, threshold, fingerprint, limitations, commit, evaluation

Example POST /ask
Request: {"query":"Who built offline mode?"}
Response includes match.id = lazarus-simboya, noMatch = false, introduction from the dataset.

Empty query → 422. Phone timeout 8s. Offline fallback is exact identity lookup, not a guessed TF-IDF score stored on the device.

7. Testing evidence

Automated tests live in `ai-api/tests/`: exact name, alias, role/feature, no-match, missing question, empty dataset, API schema, 404. Run `pytest -q` and `python build_index.py`.

Evaluation queries: at least three per member (name/alias, role, feature) plus six deliberate no-match questions including weather, president, pizza, rocket, gibberish and a cooking question about potatoes that must not pick a member.

Device matrix and tester findings are recorded in TEST_LOG.pdf and FEEDBACK.pdf. At packaging time the APK was not yet installed on physical phones; those rows are completed when Feras and Lazarus run the two-device check. Two reproducible fixes should be linked to GitHub issues after device testing (planned: API URL hint on first offline launch, and suggestion chips wrapping on small screens).

Honesty rule: no farmer, cooperative statement or impact percentage is written here because none was collected.

8. GitHub collaboration

Member	Issue	Branch	Reviews
Mohammed	#1 product home	feature/product-home-github	Reviews Mojtaba #4
Nuzha	#2 matching	feature/tfidf-matching	Reviews Mohammed #1
Ehab	#3 FastAPI	feature/fastapi-contract	Reviews Nuzha #2
Mojtaba	#4 UI	feature/mobile-ui-navigation	Reviews Ehab #3
Lazarus	#5 offline	feature/offline-cache-recovery	Reviews Mojtaba #4
Feras	#6 release	feature/apk-release-evidence	Reviews Lazarus #5

Paste `docs/issues/*.md` into GitHub before development. Tag `v1.0.0` after the APK exists. Sequential ZIP handoff is documented in `handoff/`.

Contribution map: `evidence/CONTRIBUTIONS.pdf`. Signatures are the same personal marks used on the signed G03 record dated 1 October 2026.

9. Results

The repository contains a runnable FastAPI matcher and an Expo TypeScript client with the ten required behaviours. Evaluation numbers are written into `ai-api/models/evaluation.json` by `build_index.py` so this report can quote measured top-1 and no-match rejection accuracy rather than a guessed percentage.

Screenshots of the installed APK belong in `evidence/screenshots/` after the EAS build. Until then the annotated wireframes on pages 3–4 describe the implemented screens.

10. Responsible AI, lessons, next steps

- Privacy: no phones, addresses, passwords or tokens in members.json or the APK.
- Truth: profiles refuse unverified impact. GitHub placeholder usernames must be replaced by each member.
- AI assistance is declared in evidence/AI_USE.md. Members still own the defence of their files.
- Lesson: a transparent TF-IDF index is easier to defend than a hidden prompt.
- Next: deploy FastAPI to a public HTTPS URL, run EAS, film the three-minute demo, collect four consented testers.

This report is written to match the source in this package. If a later commit changes the matcher, rebuild the index and refresh page 7.